

Proof companion

Coupling-gated cortical coherence

A companion to the repository's Lean development

First coverage stage: 16 September 2026

Abstract

This document presents mathematical explanations of 12 selected Lean results from 4 source modules. It places the assumptions and the argument of each result beside an appendix generated from Lean's elaborated statements. The saved extraction also contains 1357 nested tactic records, their before/after goal states, proof dependencies, and the fields of project structures reached through statement types. The mathematical prose is an editorial draft checked against the source during preparation; Lean verifies the formal proofs, not this prose.

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1 Reading and auditing the companion

The first coverage stage follows four kinds of argument: exact and approximate gluing, the critical exponent of stationary coherent solutions, the conditional composition of the framework, and a concrete nonconstant family. The selection is explicit in `selection.json`. Each of its declarations has an extracted statement in the appendix, including implicit type and instance parameters. Definitions and hypotheses abbreviated in the mathematics are spelled out where they carry substantive content.

The prose groups rewrites and elementary algebra into mathematical steps. Full tactic histories remain in the JSON snapshots under `data/`; Section 5.1 displays a small actual trace. A nested tactic record is an elaboration event, so its count includes wrappers and macro expansions. It is not a count of independent mathematical steps.

Three distinctions guide the reading. A theorem can depend on physical conditions supplied as structure fields. A formally valid conditional theorem can have hypotheses that require separate witnesses. A dependency on a named definition does not establish an empirical interpretation of that definition. The examples below keep these distinctions visible.

The extractor rejects elaboration errors and checks that each selected result's axioms are drawn only from `propext`, `Classical.choice`, and `Quot.sound`. This is a check of logical dependencies. It does not remove the hypotheses in the statement or the obligations carried by its structures.

2 Exact and approximate gluing

2.1 The sheaf used by the development

Let X be a topological space with the measurable and Borel structure required by the formal statement. Let F_0 assign to an open subset its finite spatial measures, with restriction maps. The Lean definition `probabilityPresheaf` denotes the sheafification F of F_0 . Despite that identifier, the initial local data are unnormalised finite measures; mass one is not part of this definition.

Claim 2.1. F is a sheaf.

Proof. Sheafification returns a sheaf together with its sheaf-property proof. Projecting that proof establishes the claim for F . $\square$

Extracted Lean statement: Appendix A.1.

Scope. This result uses the sheafification construction. It supplies no physical mechanism for producing compatible local data.

2.2 Compatible local sections have a unique extension

Claim 2.2. Let F be any sheaf of types on X , and let $(U_i)_{i \in I}$ be an open cover of X . Suppose $s_i \in F(U_i)$ and

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j} \quad (i, j \in I).$$

Then exactly one $g \in F(X)$ satisfies $g|_{U_i} = s_i$ for every i .

Proof. The unique-gluing form of the sheaf condition gives a section on $\bigcup_i U_i$, with the specified restrictions, unique among sections there. The cover equality identifies that union with X . Transport the section along this equality. Functoriality of the restriction maps shows that its restriction to each U_i is s_i .

Conversely, transport any competing section on X back to the union. Its restrictions are again the s_i , so the uniqueness supplied by the sheaf condition identifies it with the first section. Transporting forward gives the required equality on X . In the Lean proof, equality of the relevant inclusion maps uses the fact that morphisms between opens form a subsingleton. $\square$

Extracted Lean statement: Appendix A.2.

Claim 2.3. *The same conclusion holds for the development's sheafified spatial measures.*

Proof. Apply the preceding claim to $F = \text{probabilityPresheaf}$ and use the sheaf property established above. The compatible family and cover equality are exactly the hypotheses of this application. $\square$

Extracted Lean statement: Appendix A.3.

Scope. This proof is a direct term applying a previous theorem. It has no tactic history; the extracted statement and proof dependencies still cover it.

2.3 What equilibrium contributes

A thermodynamic cover supplies local sections s_i , phases θ_i , and a finite, symmetric, strictly positive coupling matrix. Its structure includes two substantive conditions: the phase configuration globally minimises the reduced Kuramoto potential, and

$$\cos(\theta_i - \theta_j) = 1 \implies s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}.$$

The second condition is an obligation on the supplied local data.

Claim 2.4. *A thermodynamic cover determines a unique global section with its prescribed local restrictions.*

Proof. The theorem characterising global minima of the reduced Kuramoto potential implies phase locking for the cover: $\cos(\theta_i - \theta_j) = 1$ for every pair. The overlap-agreement field then makes the local sections compatible. Apply the spatial-measure gluing theorem to the cover and that compatible family. $\square$

Extracted Lean statement: Appendix A.4.

Scope. This argument consumes the supplied equilibrium and overlap conditions. The existence of a physical system satisfying them is a separate matter.

2.4 Approximate agreement gives a controlled selection

Let the site set and patch index set be finite. Let $s_i(x) \geq 0$ be local mass profiles represented as functions on the ambient sites. Let $w_i(x) \geq 0$ satisfy $w_i(x) = 0$ outside U_i and $\sum_i w_i(x) = 1$ at every site. Assume that for $x \in U_i \cap U_j$, $|s_i(x) - s_j(x)| \leq \varepsilon$, where $\varepsilon \geq 0$.

Claim 2.5. *The selected profile $g(x) = \sum_j w_j(x)s_j(x)$ satisfies $\sup_{x \in U_i} |g(x) - s_i(x)| \leq \varepsilon$ for every patch i .*

Proof. Fix $x \in U_i$. Every nonzero weight $w_j(x)$ has $x \in U_j$, so its profile is within ε of $s_i(x)$. Normalisation and the triangle inequality therefore give

$$|g(x) - s_i(x)| = \left| \sum_j w_j(x)(s_j(x) - s_i(x)) \right| \leq \sum_j w_j(x)\varepsilon = \varepsilon.$$

The terms with zero weight contribute zero. The uniform metric on the finite patch converts these pointwise bounds into the displayed profile-distance bound. Lean isolates the weighted inequality in its helper `weighted_dist_le` and then applies it at each site. $\square$

Extracted Lean statement: Appendix A.5.

Scope. The profiles need not have total mass one. The result provides an approximation in the specified uniform mass metric, and assumes the overlap bounds and the subordinate normalised weights.

3 The stationary critical exponent

Write

$$E(a) = \frac{\int_0^{2\pi} \sin^2 \theta e^{a \cos \theta} d\theta}{\int_0^{2\pi} e^{a \cos \theta} d\theta}.$$

This is `vonMisesSRatio`. Its denominator is strictly positive.

3.1 The second-order expansion

Claim 3.1. As $a \rightarrow 0$,

$$E(a) = \frac{1}{2} - \frac{a^2}{16} + o(a^2).$$

Proof. The previously proved differentiability results give $E \in C^2$, $E(0) = 1/2$, $E'(0) = 0$, and $E''(0) = -1/8$. Taylor's theorem with a little- o remainder yields $E(a) = E(0) + E'(0)a + E''(0)a^2/2 + o(a^2)$. Substituting those values gives the claimed polynomial. The Lean proof applies the general Taylor theorem, identifies its second iterated derivative using the derivative lemmas, and simplifies its Taylor polynomial. $\square$

Extracted Lean statement: Appendix A.6.

Scope. The asserted remainder is $o(a^2)$. This statement supplies no fourth-order remainder estimate.

3.2 An arbitrary family approaching threshold

Claim 3.2. Let $D > 0$, let ℓ be a filter on an index type, and let K_x, r_x be real-valued functions on that type. Suppose $K_x \rightarrow 2D$, $r_x \rightarrow 0$ along ℓ , and eventually

$$K_x > 2D, \quad r_x > 0, \quad r_x = \text{selfConsistency}(K_x, D, r_x).$$

Then $r_x^2/(K_x - 2D) \rightarrow 1/D$ along ℓ .

Proof. Set $a_x = K_x r_x / D$ and write

$$E(a) = \frac{1}{2} - \frac{a^2}{16} + a^2 q(a).$$

The previous expansion gives $q(a) \rightarrow 0$ as $a \rightarrow 0$, taking the quotient with Lean's totalised division at zero. The assumed limits imply $a_x \rightarrow 0$, hence $q(a_x) \rightarrow 0$.

For an index satisfying the eventual hypotheses, positivity gives $a_x \neq 0$ and $K_x - 2D \neq 0$. The previously proved reformulation of the self-consistency equation gives $E(a_x) = D/K_x$. Substituting the expansion and rearranging yields

$$K_x - 2D = K_x a_x^2 (1/8 - 2q(a_x)), \quad \frac{r_x^2}{K_x - 2D} = \frac{D^2}{K_x^3 (1/8 - 2q(a_x))}.$$

The denominator in the last expression tends to $(2D)^3/8 = D^3 > 0$. The quotient therefore tends to $D^2/D^3 = 1/D$. The equality of the quotients holds eventually, which transfers this limit to the desired expression. $\square$

Extracted Lean statement: Appendix A.7.

Scope. Convergence to zero and eventual stationarity are hypotheses. This general statement allows the bottom filter, where its eventual and convergence conditions are vacuous. A concrete branch supplies the substantive application below. The claim establishes stationary asymptotics, with no time evolution or stability conclusion.

3.3 A branch that satisfies the hypotheses

For $D > 0$ and $K > 2D$, define $r_*(D, K)$ by choosing the coherent solution from the theorem that proves existence and uniqueness in $(0, 1]$. The Lean function `coherentBranch` uses this choice in that regime and returns zero outside it.

Claim 3.3. *If $D > 0$ and $K > 2D$, then $0 < r_*(D, K) \leq 1$ and*

$$r_*(D, K) = \text{selfConsistency}(K, D, r_*(D, K)).$$

Proof. The two inequalities on D, K select the positive branch of the definition. The specification of the chosen witness is exactly positivity, the bound by one, and the fixed-point equation supplied by the existence theorem. $\square$

Extracted Lean statement: Appendix A.8.

Claim 3.4. *For fixed $D > 0$, as K decreases to $2D$ through $K > 2D$,*

$$\frac{r_*(D, K)^2}{K - 2D} \longrightarrow \frac{1}{D}.$$

Proof. Apply the family theorem to the right-hand neighbourhood filter of $2D$, with the identity function for K . The previously proved `coherentBranch_tendsto_zero` supplies the limit of r_* at the threshold. Membership in the right-hand neighbourhood gives $K > 2D$ eventually, and the branch specification supplies positivity and stationarity there. These are all the hypotheses of the family theorem. $\square$

Extracted Lean statement: Appendix A.9.

Scope. The approach filter on the real line is nontrivial, and the branch provides actual coherent solutions. This removes the vacuity issue in this application. The conclusion still concerns stationary solutions.

4 What the conditional chain composes

The chain uses a common topological and measured substrate, its sheaf of global sections, a supplied complete metric on those sections, a neural field, a thermodynamic cover T , and a reflexive boundary with prediction map P . The precise type and instance parameters are printed in the appendix.

4.1 The common downstream argument

Suppose a sequence of energies coarse-grains to L , the elapsed parameter τ is positive, and the following named implications hold:

- $E56$: coarse-graining implies $D > 0$, $K \geq 0$, $L = K$, and an identification of the supplied field with the parameters K, D .
- $E67$: those scalar conditions imply $K > 2D$.
- $E78$: a coherent order parameter implies that the supplied cover is reached by the specified relaxation at a coupling at least K .
- $E89$: that reachability supplies a section s which restricts to the local data of T , is fixed by P , and comes with the stated Lipschitz bound on P at $\text{resonanceRate}(K, D, \tau)$.

Claim 4.1. *There is a global section compatible with a thermodynamic cover which is the unique fixed point of the prediction map.*

Proof. Apply $E56$ to the coarse-graining hypothesis and $E67$ to its scalar conclusion. The supercritical existence theorem supplies a coherent order parameter. Apply $E78$ and then $E89$ to obtain the section s , its local restrictions, the Lipschitz bound, and $P(s) = s$.

The section s makes the space of global sections nonempty. The theorem `self_of_coherent_order_parameter`, using completeness, the Lipschitz bound, coherence, and $\tau > 0$, applies the contraction argument to obtain a unique fixed point p . Since s is fixed, uniqueness gives $s = p$. Every fixed point t equals p and hence s . Pair s with the supplied cover T and its known restriction equations. This is exactly the predicate `UnifiedSelf`. $\square$

Extracted Lean statement: Appendix A.10.

Scope. The identity between the glued section and a fixed section is supplied in $E89$; the contraction argument proves uniqueness. The concrete-field conjunct of $E56$ is retained as an input constraint but is not used by this proof's downstream fixed-point argument. The source also establishes a local `Unity` fact which the concluding construction does not reference. Its constants can still occur in the unnormalised proof term; a syntactic dependency list alone therefore does not identify minimal premises.

4.2 The passive composition

The passive chain additionally supplies a finite nonempty register, its statistical-mechanics instance and update, a field and vacuum set, and measurable carrier types for predictive information. Its remaining named implications are:

- $E12$: the capacity bound implies that the field leaves the vacuum set.
- $E23$: leaving that set implies the register update is not surjective.
- $E34$: dissipation supplies a predictive-dissipation structure whose thermal scale and dissipated work match the register's temperature and heat, and whose nonpredictive information is strictly positive.

- *E45*: the predictive bound implies the chosen coarse-graining statement.

Claim 4.2. *Under $E12$ through $E89$ and the stated structural hypotheses, the passive chain concludes $UnifiedSelf$ for the supplied reflexive boundary.*

Proof. Finiteness gives the capacity bound. Apply $E12$ to obtain vacuum departure, then $E23$ to obtain a nonsurjective update. The proved dissipation theorem, using the supplied statistical-mechanics structure, gives positive heat. Apply $E34$ and retain its predictive-dissipation witness R . The bound associated with R gives the predictive node. Apply $E45$ to obtain coarse-graining and then invoke the downstream claim with $E56$ through $E89$. $\square$

Extracted Lean statement: Appendix A.11.

Scope. The eight named implications remain hypotheses. Conditions inside the statistical-mechanics, thermodynamic-cover, and other structures are additional obligations. The equalities and positivity in $E34$ constrain which witness is admitted, while the passage through $E45$ uses its predictive structure. The conclusion has no predicate asserting experience or biological implementation.

5 A concrete nonconstant family and its trace

On the three-site type with sites left, mid, and right, the development defines the nonnegative mass profile

$$d_t = (\max(1 + \cos t, 0), 1, \max(2 + \cos t, 0)).$$

The use of $\max$ expresses $\text{Real}.toNNReal$; these are nonnegative spatial masses, without a normalisation requirement.

Claim 5.1. $d_0 \neq d_\pi$.

Proof. Suppose the profiles were equal. Evaluating at the left site would equate $\max(1 + \cos 0, 0) = 2$ and $\max(1 + \cos \pi, 0) = 0$, a contradiction. $\square$

Extracted Lean statement: Appendix A.12.

Scope. This witnesses nonconstant dependence on phase for this particular family. It does not assert that every thermodynamic cover has such dependence, or that this family models a biological substrate.

5.1 Before and after the actual tactics

The following excerpt is generated from the saved elaboration data for this proof. It selects the contradiction assumption, the evaluation at the left site, and the simplification that closes the goal. The complete nested records, including wrappers sharing source spans, remain in `data/witness.json`.

```
SOURCE LINE 101: intro h
BEFORE
├ richDensity 0 ≠ richDensity Real.pi
AFTER
h : richDensity 0 = richDensity Real.pi
├ False

SOURCE LINE 102: have h1 := congrFun h Site.left
BEFORE
h : richDensity 0 = richDensity Real.pi
├ False
AFTER
```

```
h : richDensity 0 = richDensity Real.pi
hl : richDensity 0 Site.left = richDensity Real.pi Site.left
⊢ False
```

```
SOURCE LINE 103: simp [richDensity] at hl
BEFORE
h : richDensity 0 = richDensity Real.pi
hl : richDensity 0 Site.left = richDensity Real.pi Site.left
⊢ False
AFTER
No goals remain.
```

A Extracted statements and direct project dependencies

The following text is generated from elaborated Lean expressions. Universe parameter names, complete dependency lists including Mathlib, source excerpts, and goal states are retained in the JSON snapshots. Pretty-printing uses Lean's mathematical notation and can suppress implicit arguments at applications; the quantified hypotheses of the declaration remain in the displayed type. Dependency lists here show direct project constants in the proof term. They are not a minimal set of assumptions or a transitive explanation of every lemma.

A.1 sheaf-property

PhysicsOfConsciousness.probability_is_sheaf

Source, line 19; 0 nested tactic records.

```
∀ {X : TopCat} [inst : MeasurableSpace ↑X] [inst_1 : BorelSpace ↑X],
  TopCat.Presheaf.IsSheaf (PhysicsOfConsciousness.probabilityPresheaf X)
```

Direct project constants in the proof term:

PhysicsOfConsciousness.probabilityPresheaf_pre

A.2 sheaf-gluing

PhysicsOfConsciousness.sheaf_glue_unique

Source, line 52; 494 nested tactic records.

```
∀ {X : TopCat} {F : (TopologicalSpace.OPens ↑X)op ⇒ Type u},
  TopCat.Presheaf.IsSheaf F →
  ∀ {I : Type u} (cover : I → TopologicalSpace.OPens ↑X),
  iSup cover = τ →
  ∀ (s : (i : I) → F.obj (Opposite.op (cover i))),
  (∀ (i j : I),
    (CategoryTheory.ConcreteCategory.hom (F.map (CategoryTheory.homOfLE ↗).op)) (s i) =
    (CategoryTheory.ConcreteCategory.hom (F.map (CategoryTheory.homOfLE ↗).op)) (s j)) →
  ∃! g,
  ∀ (i : I),
    (CategoryTheory.ConcreteCategory.hom (F.map (CategoryTheory.homOfLE ↗).op)) g = s i
```

Direct project constants in the proof term:

PhysicsOfConsciousness.sheaf_glue_unique.match_1

A.3 probability-gluing

PhysicsOfConsciousness.probability_glue_unique

Source, line 108; 0 nested tactic records.

```
∀ {X : TopCat} [inst : MeasurableSpace ↑X] [inst_1 : BorelSpace ↑X] {I : Type u}
  (cover : I → TopologicalSpace.OPens ↑X),
  iSup cover = τ →
  ∀ (s : (i : I) → (PhysicsOfConsciousness.probabilityPresheaf X).obj (Opposite.op (cover i))),
  (∀ (i j : I),
    (CategoryTheory.ConcreteCategory.hom
      ((PhysicsOfConsciousness.probabilityPresheaf X).map (CategoryTheory.homOfLE ↗).op))
      (s i) =
    (CategoryTheory.ConcreteCategory.hom
      ((PhysicsOfConsciousness.probabilityPresheaf X).map (CategoryTheory.homOfLE ↗).op))
      (s j)) →
  ∃! g,
  ∀ (i : I),
```

```

(CategoryTheory.ConcreteCategory.hom
  ((PhysicsOfConsciousness.probabilityPresheaf X).map
    (CategoryTheory.homOfLE  $\mapsto$ ).op))
  g =
  s i

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.probabilityPresheaf
PhysicsOfConsciousness.probability_is_sheaf
PhysicsOfConsciousness.sheaf_glue_unique

```

A.4 thermodynamic-gluing

PhysicsOfConsciousness.global_section_from_thermodynamics
[Source, line 198](#); 0 nested tactic records.

```

∀ {X : TopCat} [inst : MeasurableSpace  $\mapsto$  X] [inst_1 : BorelSpace  $\mapsto$  X]
  [inst_2 : PhysicsOfConsciousness.TriangulatedManifold  $\mapsto$  X]
  [T : PhysicsOfConsciousness.ThermodynamicCover X],
  ∃! s,
    ∀ (i : PhysicsOfConsciousness.LocalSectionSynchronization.I X),
      (CategoryTheory.ConcreteCategory.hom
        ((PhysicsOfConsciousness.probabilityPresheaf X).map (CategoryTheory.homOfLE  $\mapsto$ ).op))
        s =
        PhysicsOfConsciousness.LocalSectionSynchronization.sync_to_section i

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.LocalSectionSynchronization.I
PhysicsOfConsciousness.LocalSectionSynchronization.cover
PhysicsOfConsciousness.LocalSectionSynchronization.is_cover
PhysicsOfConsciousness.LocalSectionSynchronization.sync_to_section
PhysicsOfConsciousness.ThermodynamicCover
PhysicsOfConsciousness.ThermodynamicCover.phase_locked
PhysicsOfConsciousness.ThermodynamicCover.toLocalSectionSynchronization
PhysicsOfConsciousness.TriangulatedManifold
PhysicsOfConsciousness.overlap_agreement
PhysicsOfConsciousness.probability_glue_unique

```

A.5 approximate-gluing

PhysicsOfConsciousness.ApproximateGluing.select_close
[Source, line 372](#); 47 nested tactic records.

```

∀ {α : Type u_1} {ι : Type u_2} [inst : Fintype α] [inst_1 : Fintype ι] {U : ι → Set α}
  {w s : ι → α → NNReal} {ε : NNReal},
  PhysicsOfConsciousness.ApproximateGluing.IsPartition U w →
  PhysicsOfConsciousness.ApproximateGluing.Compatible U s ε →
  ∀ (i : ι),
    PhysicsOfConsciousness.ApproximateGluing.profileDist (U i)
      (PhysicsOfConsciousness.ApproximateGluing.select w s) (s i) ≤
      ↑ε

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.ApproximateGluing.Compatible
PhysicsOfConsciousness.ApproximateGluing.IsPartition
PhysicsOfConsciousness.ApproximateGluing.profileDist
PhysicsOfConsciousness.ApproximateGluing.profileDist_le_iff
PhysicsOfConsciousness.ApproximateGluing.select

```

A.6 second-order

PhysicsOfConsciousness.vonMisesSRatio_second_order

[Source, line 196](#); 200 nested tactic records.

```
(fun a => PhysicsOfConsciousness.vonMisesSRatio a - (1 / 2 - a ^ 2 / 16)) =o[ $\mathcal{N}$  0] fun a => a ^ 2
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.contDiff_two_vonMisesSRatio
PhysicsOfConsciousness.deriv_sRatioDeriv_zero
PhysicsOfConsciousness.deriv_vonMisesSRatio
PhysicsOfConsciousness.sRatioDeriv
PhysicsOfConsciousness.sRatioDeriv_zero
PhysicsOfConsciousness.vonMisesSRatio
PhysicsOfConsciousness.vonMisesSRatio_zero
```

A.7 family-exponent

PhysicsOfConsciousness.coherent_solution_critical_exponent

[Source, line 224](#); 340 nested tactic records.

```
∀ {α : Type u_1} {l : Filter α} {K r : α → ℝ} {D : ℝ},
  0 < D →
    Filter.Tendsto K l ( $\mathcal{N}$  (2 * D)) →
      Filter.Tendsto r l ( $\mathcal{N}$  0) →
        (∀f (x : α) in l,
          2 * D < K x ∧ 0 < r x ∧ r x = PhysicsOfConsciousness.selfConsistency (K x) D (r x)) →
          Filter.Tendsto (fun x => r x ^ 2 / (K x - 2 * D)) l ( $\mathcal{N}$  (1 / D))
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.coherent_iff_sRatio_eq
PhysicsOfConsciousness.selfConsistency
PhysicsOfConsciousness.vonMisesSRatio
PhysicsOfConsciousness.vonMisesSRatio_second_order
```

A.8 branch-spec

PhysicsOfConsciousness.coherentBranch_spec

[Source, line 288](#); 84 nested tactic records.

```
∀ {D K : ℝ},
  0 < D →
    2 * D < K →
      0 < PhysicsOfConsciousness.coherentBranch D K ∧
        PhysicsOfConsciousness.coherentBranch D K ≤ 1 ∧
          PhysicsOfConsciousness.coherentBranch D K =
            PhysicsOfConsciousness.selfConsistency K D (PhysicsOfConsciousness.coherentBranch D K)
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.coherentBranch
PhysicsOfConsciousness.coherentBranch._proof_1
PhysicsOfConsciousness.coherentBranch.eq_1
PhysicsOfConsciousness.critical_coupling
PhysicsOfConsciousness.critical_coupling._proof_1
PhysicsOfConsciousness.critical_coupling.eq_1
PhysicsOfConsciousness.selfConsistency
PhysicsOfConsciousness.supercritical_fixed_point_existsUnique
```

A.9 branch-exponent

PhysicsOfConsciousness.coherentBranch_critical_exponent
[Source, line 317](#); 16 nested tactic records.

```
∀ {D : ℝ},
  0 < D →
  Filter.Tendsto (fun K => PhysicsOfConsciousness.coherentBranch D K ^ 2 / (K - 2 * D))
    (ℳ[>] (2 * D)) (ℳ (1 / D))
```

Direct project constants in the proof term:

PhysicsOfConsciousness.coherentBranch
PhysicsOfConsciousness.coherentBranch_spec
PhysicsOfConsciousness.coherentBranch_tendsto_zero
PhysicsOfConsciousness.coherent_solution_critical_exponent
PhysicsOfConsciousness.selfConsistency

A.10 downstream-chain

PhysicsOfConsciousness.Chain.chain_from_coarseGrains
[Source, line 796](#); 91 nested tactic records.

```
∀ {X : TopCat} [inst : MeasureTheory.MeasureSpace ↑X] [inst_1 : BorelSpace ↑X]
  [inst_2 : PhysicsOfConsciousness.TriangulatedManifold ↑X]
  [inst_3 : MetricSpace PhysicsOfConsciousness.GlobalSection]
  [CompleteSpace PhysicsOfConsciousness.GlobalSection]
  (kernel : PhysicsOfConsciousness.StochasticNeuralField ↑X) {E : ℕ → ℝ} {L K D τ : ℝ},
  PhysicsOfConsciousness.Chain.CoarseGrains E L →
  0 < τ →
  ∀ (T : PhysicsOfConsciousness.ThermodynamicCover X)
    (rb : PhysicsOfConsciousness.ReflexiveBoundary X),
    PhysicsOfConsciousness.Chain.E56 kernel E L K D →
    PhysicsOfConsciousness.Chain.E67 L K D →
    PhysicsOfConsciousness.Chain.E78 K D T →
    PhysicsOfConsciousness.Chain.E89 K D τ T rb →
    PhysicsOfConsciousness.Chain.UnifiedSelf rb
```

Direct project constants in the proof term:

PhysicsOfConsciousness.Chain.CoarseGrains
PhysicsOfConsciousness.Chain.Coherent
PhysicsOfConsciousness.Chain.E56
PhysicsOfConsciousness.Chain.E67
PhysicsOfConsciousness.Chain.E78
PhysicsOfConsciousness.Chain.E89
PhysicsOfConsciousness.Chain.FieldRealizes
PhysicsOfConsciousness.Chain.IsUnifiedBy
PhysicsOfConsciousness.Chain.Self
PhysicsOfConsciousness.Chain.UnifiedSelf
PhysicsOfConsciousness.Chain.Unity
PhysicsOfConsciousness.Chain.coherent_of_supercritical
PhysicsOfConsciousness.Chain.self_of_coherent_order_parameter
PhysicsOfConsciousness.Chain.unity_of_cover
PhysicsOfConsciousness.GlobalSection
PhysicsOfConsciousness.IsEMFieldCoupling
PhysicsOfConsciousness.ReflexiveBoundary
PhysicsOfConsciousness.ReflexiveBoundary.predict
PhysicsOfConsciousness.StochasticNeuralField
PhysicsOfConsciousness.ThermodynamicCover
PhysicsOfConsciousness.ThermodynamicCover.IsReachedByRelaxation
PhysicsOfConsciousness.TriangulatedManifold
PhysicsOfConsciousness.resonanceRate

A.11 passive-chain

PhysicsOfConsciousness.Chain.chain

Source, line 858; 66 nested tactic records.

```
∀ {X : TopCat} [inst : MeasureTheory.MeasureSpace ↑X] [inst_1 : BorelSpace ↑X]
  [inst_2 : PhysicsOfConsciousness.TriangulatedManifold ↑X]
  [inst_3 : MetricSpace PhysicsOfConsciousness.GlobalSection]
  [CompleteSpace PhysicsOfConsciousness.GlobalSection]
  (kernel : PhysicsOfConsciousness.StochasticNeuralField ↑X) {sys : Type u_1} [inst_5 : Fintype sys]
  [inst_6 : DecidableEq sys] [Nonempty sys]
  [inst_8 : PhysicsOfConsciousness.StatisticalMechanics sys] (upd : sys → sys) {Y : Type u_2}
  {V : Type u_3} (vac : Set V) (phi : Y → V) (Xs : Type u_4) (Sg : Type u_5) (Sg' : Type u_6)
  [inst_9 : MeasurableSpace Xs] [inst_10 : MeasurableSpace Sg] [inst_11 : MeasurableSpace Sg']
  (E : ℕ → ℝ) (L : ℝ) {K D τ : ℝ},
  0 < τ →
  ∀ (T : PhysicsOfConsciousness.ThermodynamicCover X)
    (rb : PhysicsOfConsciousness.ReflexiveBoundary X),
    PhysicsOfConsciousness.Chain.E12 sys vac phi →
    PhysicsOfConsciousness.Chain.E23 vac phi upd →
    PhysicsOfConsciousness.Chain.E34 upd Xs Sg Sg' →
    PhysicsOfConsciousness.Chain.E45 Xs Sg Sg' E L →
    PhysicsOfConsciousness.Chain.E56 kernel E L K D →
    PhysicsOfConsciousness.Chain.E67 L K D →
    PhysicsOfConsciousness.Chain.E78 K D T →
    PhysicsOfConsciousness.Chain.E89 K D τ T rb →
    PhysicsOfConsciousness.Chain.UnifiedSelf rb
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.Capacity
PhysicsOfConsciousness.Chain.CoarseGrains
PhysicsOfConsciousness.Chain.Dissipates
PhysicsOfConsciousness.Chain.E12
PhysicsOfConsciousness.Chain.E23
PhysicsOfConsciousness.Chain.E34
PhysicsOfConsciousness.Chain.E45
PhysicsOfConsciousness.Chain.E56
PhysicsOfConsciousness.Chain.E67
PhysicsOfConsciousness.Chain.E78
PhysicsOfConsciousness.Chain.E89
PhysicsOfConsciousness.Chain.LeavesVacuum
PhysicsOfConsciousness.Chain.PredictiveBound
PhysicsOfConsciousness.Chain.UnifiedSelf
PhysicsOfConsciousness.Chain.capacity
PhysicsOfConsciousness.Chain.chain_from_coarseGrains
PhysicsOfConsciousness.Chain.dissipates_of_not_surjective
PhysicsOfConsciousness.Chain.predictiveBound_of_nonempty
PhysicsOfConsciousness.GlobalSection
PhysicsOfConsciousness.PredictiveDissipation
PhysicsOfConsciousness.PredictiveDissipation.dissipatedWork
PhysicsOfConsciousness.PredictiveDissipation.nonpredictive
PhysicsOfConsciousness.PredictiveDissipation.thermalEnergy
PhysicsOfConsciousness.ReflexiveBoundary
PhysicsOfConsciousness.StatisticalMechanics
PhysicsOfConsciousness.StatisticalMechanics.toThermodynamics
PhysicsOfConsciousness.StochasticNeuralField
PhysicsOfConsciousness.ThermodynamicCover
PhysicsOfConsciousness.Thermodynamics.temperature
PhysicsOfConsciousness.TriangulatedManifold
PhysicsOfConsciousness.heat_dissipation
```

A.12 nonconstant-witness

PhysicsOfConsciousness.Examples.richDensity_not_const

Source, line 100; 19 nested tactic records.

```
PhysicsOfConsciousness.Examples.richDensity 0 ≠ PhysicsOfConsciousness.Examples.richDensity Real.pi
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Examples.Site  
PhysicsOfConsciousness.Examples.Site.left  
PhysicsOfConsciousness.Examples.richDensity
```

B Two extracted physical obligations

These projection types are extracted from the structures used by the gluing results. They show the equilibrium and overlap-agreement obligations described in Section 2. The snapshots contain the other encountered structure fields as well. The exporter follows project statement types and parent fields; it does not claim a complete semantic expansion through every definition body or every Mathlib structure.

```
PhysicsOfConsciousness.ThermodynamicCover.thermodynamic_equilibrium  
∀ {X : TopCat} {inst : MeasurableSpace ↑X} {inst_1 : BorelSpace ↑X}  
  {inst_2 : PhysicsOfConsciousness.TriangulatedManifold ↑X}  
  [self : PhysicsOfConsciousness.ThermodynamicCover X]  
  (theta : PhysicsOfConsciousness.LocalSectionSynchronization.I X → ℝ),  
PhysicsOfConsciousness.kuramoto_potential_dynamic  
  { omega := fun x => 0, A := PhysicsOfConsciousness.ThermodynamicCover.A, symm := ... }  
  PhysicsOfConsciousness.LocalSectionSynchronization.phase ≤  
PhysicsOfConsciousness.kuramoto_potential_dynamic  
  { omega := fun x => 0, A := PhysicsOfConsciousness.ThermodynamicCover.A, symm := ... } theta  
  
PhysicsOfConsciousness.LocalSectionSynchronization.section_agrees_of_phase_eq  
∀ {X : TopCat} {inst : MeasurableSpace ↑X} {inst_1 : BorelSpace ↑X}  
  {inst_2 : PhysicsOfConsciousness.TriangulatedManifold ↑X}  
  [self : PhysicsOfConsciousness.LocalSectionSynchronization X]  
  (i j : PhysicsOfConsciousness.LocalSectionSynchronization.I X),  
Real.cos  
  (PhysicsOfConsciousness.LocalSectionSynchronization.phase i -  
    PhysicsOfConsciousness.LocalSectionSynchronization.phase j) =  
1 →  
(CategoryTheory.ConcreteCategory.hom  
  ((PhysicsOfConsciousness.probabilityPresheaf X).map (CategoryTheory.homOfLE ...).op))  
  (PhysicsOfConsciousness.LocalSectionSynchronization.sync_to_section i) =  
(CategoryTheory.ConcreteCategory.hom  
  ((PhysicsOfConsciousness.probabilityPresheaf X).map (CategoryTheory.homOfLE ...).op))  
  (PhysicsOfConsciousness.LocalSectionSynchronization.sync_to_section j)
```